Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*		**Indicative content:** 1. Wavelength at peak intensity and Wien's Law [can be used to] determine [surface] temperature of the star. 2 Power emitted per square meter or area can be calculated using Stefan's law [details not required] 3. Colour of star can be deduced from wavelength of max intensity / spectrum. 4. Line **absorption** spectrum shown. 5 Line spectrum arises from passage of [continuous] spectrum/light/radiation / photons through stellar atmosphere 6 absorbing atoms/elements can be identified from wavelength of lines **Other relevant points:** • Total area under graph represents total power radiated • Absorption spectrum indicates temperature and generation of star • Redshift gives radial / recessional/ velocity/ distance • Reference to inverse square law and distance	6			6		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			5-6 marks At least 5 relevant points given *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** At least 3 relevant points given *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.* **1-2 marks** At least 1 relevant points given *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** *No attempt made or no response worthy of credit.*						

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	Recall of $I = P \div 4\pi R^2$ in any form (1) Substitution: P (or Luminosity) = $1.32 \times 10^{-8} \times 4\pi \times (1.58 \times 10^{17})^2$ (1) $P = 4.1(4) \times 10^{27}$ W seen (1)	1	1 1		3	3	
		(ii)	Substitution: $P = A\sigma T^4$ in any form e.g.: $4 \times 10^{27} = A \times 5.67 \times 10^{-8} \times (7\,700)^4$ (1) [Accept $4.1(4)10^{27}$ for P] $A = 2.0 \times 10^{19}$ m^2 (1) [2.1×10^{19} if 4.14×10^{27} used] Diameter = 2.5×10^9 m (1) [2.6×10^9 if 2.1×10^{19} used] Use of πR^2 rather than $4\pi R^2 \rightarrow 5 \times 10^9$ m (2 marks)	1 1	1		3	3	
			Question 6 total	**9**	**3**	**0**	**12**	**6**	**0**